Zero-State Edge-Distribution Recursion for Dynamic Graph and Spectral Estimation

Gurui Jin and Ercan E. Kuruoğlu

***Abstract*—Dynamic graph tracking requires estimating both the existence and strength of edges. We study a finite edge-distribution recursion with an explicit zero state. Node dynamics yield an edge-weight regression; Markov prediction and mean-field updating maintain edge probabilities. A support decision followed by conditional amplitude estimation produces a Laplacian and its complete spectrum. On 24 synthetic condition trajectories comprising 1,728 blocks, mean relative edge-weight and eigenvalue errors decrease from 0.15134 and 0.08741 for a task-adapted hard-threshold sparse EKF to 0.12163 and 0.06527. False splits of connected graphs decrease from 76/1068 to 8/1068. These improvements come with greater computation time and worse event-block increment error. We evaluate this tradeoff through the structural, weight, and spectral outputs of a single estimator.**

***Index Terms*—Dynamic graph learning, edge-weight estimation, recursive inference, mean-field approximation, graph spectra.**

I. Introduction

GRAPH signal processing describes interactions between nodes and uses Laplacian spectra to characterize signal variation [1]. When edges disappear, reappear, and change strength, online inference must distinguish absent edges from weak connections while tracking their amplitudes. We ask how observed node trajectories, under a specified interaction model, can provide recursive edge estimates and consistent spectral outputs U_t, Λ_t.

Natali *et al.* study time-varying optimization for online graph learning [2]. Dabush *et al.* treat edge weights as latent states and graph-filtered signals as observations in a sparsity-aware EKF [3]. Edge-weight Laplacian parameterization and the linear special case of first-order graph filtering are established foundations. Our coordinate update likewise uses standard mean-field variational inference [4].

Our central design maintains probabilities for both zero weight and multiple positive amplitudes in one recursive edge distribution, then uses that distribution for support and weight estimation. The estimator combines finite-state prediction, factorized likelihood updating, and support readout. The zero state represents edge absence; the positive grid accommodates weighted edges; factorization controls the number of joint states. We test the estimation–cost tradeoff of this specific design against literature-derived baselines, rather than claim the regression identity or mean-field principle as a new theory.

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II. From Node Observations to Edge Information

A. Unknowns, observations, and modeling rationale

Let $s_{t,k} \in \mathbb{R}^n$ contain scalar node signals. Index t denotes a graph-update block and k a within-block sample. The unknown $\theta_t \in [0,1]^m$ contains candidate *edge* weights, not node values. Each column of $B = [b_1, \ldots, b_m]$ has $+1, -1$ at the endpoints of one edge and zeros elsewhere, giving

$$L(\theta_t) = B\,\mathrm{diag}(\theta_t)B^{\mathsf{T}}. \tag{1}$$

The arbitrary orientation does not affect $b_e b_e^{\mathsf{T}}$. We use the first-order local interaction model

$$s_{t,k+1} = (aI - \eta L(\theta_t))s_{t,k} + u_{t,k} + \epsilon_{t,k}. \tag{2}$$

At node i, the interaction is $-\eta \sum_j \theta_{ij}(s_{i,k} - s_{j,k})$, added to self-evolution $as_{i,k}$ and known input $u_{i,k}$. This discrete diffusion-type graph filter [1] applies when signals follow such coupling. All states are synchronously observed; a, η, u, σ are known. Process innovations are independent across transitions, independent of the current state, and distributed as $\mathcal{N}(0, \sigma^2 I)$. There is no additional sensor measurement noise.

For intuition, consider room temperatures coupled by heat exchange. Edge weights describe exchange strengths, while $as + u$ predicts the next temperatures without inter-room exchange. The difference from the observed next temperatures contains coupling and disturbance. Repeated temperature changes therefore inform the unknown strengths. This analogy explains the model; the experiments remain synthetic.

B. Residual meaning and derivation

The regression response is computed from observations:

$$r_{t,k} = as_{t,k} + u_{t,k} - s_{t,k+1}. \tag{3}$$

It is the difference between a prediction omitting graph interactions and the observed next state, not an additional sensor or pure noise. Suppressing indices, substitution gives

$$\begin{aligned} r &= as + u - [(aI - \eta L)s + u + \epsilon] \\ &= \eta L s - \epsilon \\ &= \eta B \operatorname{diag}(\theta) B^{\mathsf{T}} s - \epsilon \\ &= \underbrace{\eta B \operatorname{diag}(B^{\mathsf{T}} s)}_{H} \theta - \epsilon. \end{aligned} \tag{4}$$

The last step uses $\operatorname{diag}(\theta)d = \operatorname{diag}(d)\theta$ for $d = B^{\mathsf{T}}s$. Column e is $\eta b_e(b_e^{\mathsf{T}}s)$: an endpoint indicator scaled by the endpoint signal difference and coupling coefficient. Once s is observed, H is known and the equation is linear in θ.

Before observation, the innovation and unknown weights make the next state and residual random. After observation, the residual is numerically determined. These statements are compatible: the likelihood describes how the observed value depends on unknown weights. Arbitrary node values alone do not identify a graph without an interaction model.

C. Three-node example and block stacking

For candidate edges $(1,2),(1,3),(2,3)$, take $s = (1,0,0.5)^{\mathsf{T}}$, $a = 0.9$, $\eta = 0.1$, and $u = 0$. Then

$$B = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & -1 \end{bmatrix}, \quad H = \begin{bmatrix} .1 & .05 & 0 \\ -.1 & 0 & -.05 \\ 0 & -.05 & .05 \end{bmatrix}. \tag{5}$$

For example, the second column is $0.1(1,0,-1)^{\mathsf{T}}(1 - 0.5)$. Generating a noiseless example with $\theta = (0.8, 0.5, 0)^{\mathsf{T}}$ gives $H\theta = (.105, -.08, -.025)^{\mathsf{T}}$ and $s_{k+1} = (.795, .08, .475)^{\mathsf{T}}$. An observer computes the same residual as $0.9s - s_{k+1}$ without knowing the generating weights. This verifies the identity, not unique recovery from one observation.

Keeping θ_t fixed for K transitions provides K equations. Stacking places them vertically, without averaging:

$$\underbrace{\begin{bmatrix} r_{t,0} \\ \vdots \\ r_{t,K-1} \end{bmatrix}}_{r_t \in \mathbb{R}^{nK}} = \underbrace{\begin{bmatrix} H_{t,0} \\ \vdots \\ H_{t,K-1} \end{bmatrix}}_{H_t \in \mathbb{R}^{nK \times m}} \theta_t - \epsilon_t. \tag{6}$$

Each transition adds constraints on the same weights. Equal endpoint signals give $B^{\mathsf{T}}s = 0$ and $H = 0$, conveying no instantaneous edge information. Full column rank of the stacked design suffices for uniqueness of unconstrained noiseless regression; a prior does not replace missing observation information.

III. Zero-State Edge-Distribution Recursion

A. Finite representation and prediction

Each edge takes values $v_j = j/64$, $j = 0, \ldots, 64$, with approximate probabilities $q_{t,e}(j)$. The zero state denotes an absent edge. Prediction is

$$p^-_{t,e}(j) = \sum_{i=0}^{64} q_{t-1,e}(i) T_{ij}, \tag{7}$$

where T_{ij} describes a transition between weight states. The experiments fix zero-to-positive and positive-to-zero switching probabilities to 0.01 per block, using a discretized Gaussian transition for positive amplitudes. The first block uses the initial distribution with zero-state probability 0.5; subsequent blocks each perform one prediction.

B. Why mean-field, and how it updates

Exact enumeration requires 65^m joint configurations. The factorization $q_t(\theta_t) = \prod_e q_{t,e}(\theta_{t,e})$ stores $65m$ probabilities. Interactions enter each update through current means, trading a general representation of inter-edge dependence for tractable computation [4].

Multiplying the conditional transition densities along the observed trajectory yields the log likelihood

$$\begin{aligned} \ell_t(\theta_t) &= -\tfrac{1}{2}\theta_t^{\mathsf{T}} G_t \theta_t + h_t^{\mathsf{T}} \theta_t + C, \\ G_t = H_t^{\mathsf{T}} H_t / \sigma^2, &\quad h_t = H_t^{\mathsf{T}} r_t / \sigma^2. \end{aligned} \tag{8}$$

This sequential likelihood does not assume that the entire trajectory-dependent design is independent of all process innovations. Let $\mu_{t,f} = \sum_j v_j q_{t,f}(j)$. The standard rule $\log q_{t,e} = \log p^-_{t,e} + \mathbb{E}_{-e}\ell_t + \text{const}$ gives

$$\begin{aligned} q_{t,e}(j) &\propto p^-_{t,e}(j) \exp[-\tfrac{1}{2} G_{t,ee} v_j^2 + c_{t,e} v_j], \\ c_{t,e} &= h_{t,e} - \sum_{f \neq e} G_{t,ef} \mu_{t,f}. \end{aligned} \tag{9}$$

The cross term $-G_{ef}\theta_e\theta_f$ becomes $-G_{ef}v_j\mu_f$ after taking the expectation; terms independent of v_j enter the normalizer. Updates use a fixed edge order, normalize each distribution, and stop when maximum mean and probability changes both fall below 10^{-8}, or after 30 sweeps. Capped blocks are retained and recorded.

C. One estimator and one point-output rule

We fix the support readout

$$\hat{\theta}_{t,e} = \begin{cases} 0, & q_{t,e}(0) \geq 1/2, \\ \dfrac{\sum_{j=1}^{64} v_j q_{t,e}(j)}{1 - q_{t,e}(0)}, & q_{t,e}(0) < 1/2. \end{cases} \tag{10}$$

This makes an existence decision followed by conditional amplitude estimation. For example, $q(0) = 0.7$ produces zero instead of a small positive mixture mean.

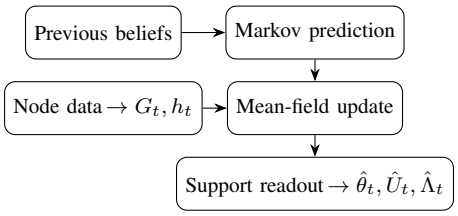


Fig. 1. Prediction, observation update, and output of the single proposed estimator. Updated beliefs carry to the next block.

The rule is neither a global joint MAP estimate nor the unconditional minimum mean-square estimator.

Equation (1) then gives $\hat{L}_t$, and a full eigendecomposition yields $\hat{L}_t = \hat{U}_t \hat{\Lambda}_t \hat{U}_t^{\mathsf{T}}$. Recursive memory resides in the edge distributions; U, Λ are spectral outputs of the point graph, computed by full EVD.

D. Relationship to existing filters

Our estimator and KF share Bayesian prediction and updating. Gaussian KF represents a continuous state by its mean and covariance; here, an explicit zero mass and positive grid probabilities are updated using (9). Gaussian process innovations do not imply a single Gaussian edge posterior. The distinction is not observation nonlinearity: (4) is exactly linear in edge weights and needs no EKF Taylor approximation. Dabush *et al.* also address more general nonlinear graph filters [3].

IV. Experiments and Existing-Method Comparisons

A. Common task and metrics

We use 6/12 nodes, two structural-change generators, two excitation conditions, and three graph draws per condition: 24 trajectories of 72 blocks, each with $K = 48$ transitions. All $m = n(n-1)/2$ candidate edges are available. Parameters are $a = 0.95$, $\eta = 0.8/n$, and $\sigma = 0.03$, with zero-mean Gaussian input standard deviation 0 or 0.3. Graph and observation seeds were fixed before testing; estimator parameters carry over from development. Paired excitation conditions share graphs, so the 1,728 blocks are not independent replicates.

Literature-derived baselines. The main comparison adapts the hard-threshold branch of the public implementation accompanying Dabush *et al.* [3], [5] to the same regression, with $[0, 1]$ constraints and the previously selected threshold 0.1. We call it the sparse EKF adaptation, distinguishing it from the paper's principal ℓ_1 algorithm. The supplementary ProbSSM baseline comes from the same authors' repository [5]. It maintains node-local binary incident-edge configurations with fixed positive amplitude 1. Its average and local-MAP readouts share inference. It is a repository baseline, not another proposed method or a substitute for the paper's main algorithm.

TABLE I
Same-observation graph estimation. Lower errors and false splits are better; more correct partitions are better.

Method	E_θ	E_λ	Partitions	False splits
Proposed	.12163	.06527	404/660	8/1068
Sparse EKF adapted	.15134	.08741	330/660	76/1068
ProbSSM average	.16894	.08844	2/660	0/1068
ProbSSM local MAP	.19677	.10135	545/660	103/1068

Every method accesses the same observed signals and known parameters, and computes a full spectrum from its point graph. For ascending eigenvalues λ_t, blockwise errors are

$$E_\theta(t) = \frac{\|\hat{\theta}_t - \theta_t\|_2}{\max(\|\theta_t\|_2, 10^{-15})}, \qquad E_\lambda(t) = \frac{\|\hat{\lambda}_t - \lambda_t\|_2}{\max(\|\lambda_t\|_2, 10^{-15})}. \tag{11}$$

We report means across blocks. Correct partition recovery requires all connected components to match on a disconnected true graph. A false split occurs when a connected true graph is estimated as disconnected. Their denominators are 660 and 1068 blocks, respectively. Components use strictly positive edges in the point graph.

B. Accuracy, change error, and cost

Relative to the sparse EKF adaptation, mean edge and eigenvalue errors decrease by about 19.6% and 25.3%, with improved aggregate partition counts. ProbSSM local MAP recovers more disconnected partitions but produces more false splits. ProbSSM was added on previously used observations; its fixed amplitude mismatches multivalued true weights, and its parameters were not independently optimized.

For blocks with true support changes, we average squared point-increment errors over all edges and event blocks, then take the square root. The proposed method scores 0.11435 versus 0.09379 for the sparse EKF adaptation: improved average estimation does not also improve abrupt-change response. Single-thread online time, including statistics, inference, and full EVD, averages 9.75 versus 1.04 seconds per trajectory. Initialization, warmup, storage, and evaluation are excluded; the machine was shared. All 14/1728 capped proposed-method blocks remain in the results. ProbSSM timing comes from another run and is not combined into speed ratios.

Under the archived common-gap selection, projection errors over 5,517 positive simple eigenvalue directions average 0.86552 (proposed) and 0.87621 (sparse

EKF adaptation). Eligibility depends on truth and all outputs of the original experiment, excludes repeated eigenspaces, and does not establish a general advantage for the complete U.

V. Conclusion and a Spectral Task Interface

A single zero-state edge-distribution recursion estimates dynamic support and amplitude and provides consistent complete spectral outputs. Synthetic comparisons show lower mean weight and eigenvalue errors and fewer false splits than the listed sparse EKF adaptation, at greater computational cost and event-increment error.

For a graph fixed within a block, $z_k = U^{\mathsf{T}} s_k$ satisfies $z_{k+1} = (aI - \eta\Lambda)z_k + U^{\mathsf{T}}u_k + U^{\mathsf{T}}\epsilon_k$. Thus U describes collective variation modes and Λ determines their propagation factors. A UAV study must specify the physical node signals and edges, then compare equal-cost decisions using estimated spectra, true spectra, and no spectra. This paper validates the estimation layer; decision-loop benefits remain an experimental objective. Partial observations, model mismatch, and correlated posterior representations also require further study.

References

[1] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, 2013.

[2] A. Natali, E. Isufi, M. Coutino, and G. Leus, "Learning time-varying graphs from online data," *IEEE Open J. Signal Process.*, 2022, doi: 10.1109/OJSP.2022.3178901.

[3] L. Dabush, N. Shlezinger, and T. Routtenberg, "Sparsity-aware extended Kalman filter for tracking dynamic graphs," arXiv:2507.09999, 2025.

[4] D. M. Blei, A. Kucukelbir, and J. D. McAuliffe, "Variational inference: A review for statisticians," *J. Amer. Statist. Assoc.*, vol. 112, no. 518, pp. 859–877, 2017.

[5] L. Dabush, "PolyEKF," source snapshot cbb1a0a (full commit recorded with the experiments), `baseline_prob_ssm.py` and EKF implementation. [Online]. Available: https://github.com/lital-dgold/PolyEKF